

EXPERIMENT 9

Implement Connection using UDP

UDP • CONNECTIONLESS

Objective

- ✓ Understand the basic working principles and characteristics of UDP (User Datagram Protocol).
- ✓ Implement UDP client–server communication using Python socket programming.
- ✓ Establish communication between server.py and client.py using UDP sockets.
- ✓ Observe how datagrams are sent and received without a dedicated connection between client and server.

Theory

UDP (User Datagram Protocol) is a connectionless transport-layer protocol used for communication between devices on a network. Unlike TCP, it does not set up a connection before data is sent and offers no guarantee of delivery, ordering, or retransmission. Data instead travels as independent datagrams, each one carrying the destination IP address and port along with it.

In this experiment, the server creates a UDP socket and binds it to an IP address and port using `bind()`, while the client sends data straight to the server using `sendto()` (or, once "connected" for convenience, `send()`). The server reads incoming datagrams with `recvfrom()` and can reply directly to the sender. Because there is no three-way handshake or persistent session, UDP is lighter and faster than TCP — which is exactly why it suits latency-sensitive use cases such as streaming, online gaming, DNS lookups, and real-time communication.

TCP VS UDP — QUICK COMPARISON

Aspect	TCP	UDP
Connection	Connection-oriented (handshake)	Connectionless
Reliability	Guaranteed, ordered delivery	Best-effort, no guarantee
Speed	Slower (overhead of checks)	Faster, lower latency
Typical use	Web, email, file transfer	Streaming, gaming, DNS

Practical Steps

1. Open the project folder in a code editor such as VS Code, PyCharm, or IntelliJ.
2. Create two Python files named `server.py` and `client.py`.
3. Paste the server-side code into `server.py` and the client-side code into `client.py`.
4. Save both files.
5. Open two terminals (or a split terminal).
6. In Terminal 1, start the server with `python server.py`.
7. In Terminal 2, start the client with `python client.py`.
8. Exchange a few datagrams between the client and server CLIs, then type `Over` on the client side to end the session.

Code

server.py

```
import socket

server_socket = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
server_socket.bind(("127.0.0.1", 5000))
print("UDP Server started")
print("Waiting for messages...")

while True:
    message, client_address = server_socket.recvfrom(1024)
    message = message.decode()
    print("Client:", message)
    if message == "Over":
        break
    reply = input("Enter reply: ")
    server_socket.sendto(reply.encode(), client_address)

server_socket.close()
print("Server closed")
```

client.py

```
import socket

client_socket = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
client_socket.connect(("127.0.0.1", 5000))
print("UDP Client started")

while True:
    message = input("Enter message: ")
    client_socket.send(message.encode())    ## encode(): string -> bytes
    if message == "Over":
        break
    reply = client_socket.recv(1024).decode()
    print("Server:", reply)

client_socket.close()
print("Client closed")
```

Result

client.py — Terminal 2

Terminal 2 • client.py

```
PS D:\Lakshdeep\CN_Lab\UDP_Experiment> python client.py
UDP Client started
Enter message: Hello server, this is a UDP test from Lakshdeep
Server: Datagram received, no handshake needed here
Enter message: Exactly, that is the whole point of UDP
Server: Right – fast and lightweight, but no delivery guarantee
Enter message: Good for real-time stuff like gaming and DNS then
Server: Exactly that
Enter message: Over
Client closed
```

server.py — Terminal 1

Terminal 1 • server.py

```
PS D:\Lakshdeep\CN_Lab\UDP_Experiment> python server.py
UDP Server started
Waiting for messages...
Client: Hello server, this is a UDP test from Lakshdeep
Enter reply: Datagram received, no handshake needed here
Client: Exactly, that is the whole point of UDP
Enter reply: Right – fast and lightweight, but no delivery guarantee
Client: Good for real-time stuff like gaming and DNS then
Enter reply: Exactly that
Client: Over
Server closed
```

Learning Outcome

- ✓ Understood the working and characteristics of UDP and how it differs from connection-oriented protocols such as TCP.
- ✓ Built and ran a UDP client–server application in Python for sending and receiving datagrams.
- ✓ Compared UDP against TCP in terms of reliability, overhead, and typical real-world use cases.

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